Plotter enclosures · cut & bend pack

Two stacked electronics boxes (control on top, power below), each a folded body + a folded lid, plus two flat base plates. Prepared by Allwin Tom · 2026-10-11 · v3

#	File	Part	Material	Qty	Flat (mm)	Bends	Folded outside (mm)
1	`control_body.dxf`	CONTROL BOX · BODY	1.0 mm 304 stainless, brushed, film on	1	699.0 × 434.8	11	452 W × 297 D × 111 H
2	`control_lid.dxf`	CONTROL BOX · LID	1.0 mm 304 stainless, brushed, film on	1	484.5 × 423.5	6	456 W × 300.0 D × 111 H (back)
3	`control_base_plate.dxf`	CONTROL BOX · BASE PLATE	2.0 mm aluminium 5052 (or 1.5 mm 304)	1	440.0 × 285.0	0	flat, 440 × 285, notched
4	`power_body.dxf`	POWER BOX · BODY	1.0 mm 304 stainless, brushed, film on	1	819.0 × 494.8	11	452 W × 297 D × 171 H
5	`power_lid.dxf`	POWER BOX · LID	1.0 mm 304 stainless, brushed, film on	1	484.5 × 483.5	6	456 W × 300.0 D × 171 H (back)
6	`power_base_plate.dxf`	POWER BOX · BASE PLATE	2.0 mm aluminium 5052 (or 1.5 mm 304)	1	440.0 × 285.0	0	flat, 440 × 285, notched

Bending

- Every flat is drawn from the **inside face**. Every bend is **UP 90°**, inner radius **R1**.
- Flats assume K 0.44, bend allowance 2.262 mm, bend deduction 1.738 mm per 90° bend (1 mm 304, R = t). **If your bend deduction for 1 mm 304 at R1 differs, tell us the number**: we re-issue the flats so every hole lands where it should.
- Bend order: small lips first (top lips, back lips, corner tabs, base back lip), then the side walls, then the front wall last. Lid: return lips first, then the top-to-back bend.
- Corner reliefs and end-of-flange reliefs are already cut into the outline.
- **Deliberate gaps, please keep:** flange ends stop 2.5 mm short of neighbouring flanges; side-wall front edges and corner tabs stop 0.5 mm short of the front wall (the rivets close it); the lid is 1.0 mm per side larger than the body.

Layers

- **Visible** — cut (outline, holes, cutouts, vents).
- **Bend Center** (dash-dot) and **Visible Narrow** (bend-zone edges) — bend references, **do not cut**.
- **Bend Notes**, **Leaders**, **Title** — information only, do not cut or engrave.
- DXF R12, millimetres, 1:1.

Finish & handling

- 304 stainless, brushed, protective film left on. Grain the same direction on all visible faces of both boxes.
- Deburr all edges. No paint or coating.
- Base plates: 2 mm 5052 aluminium (1.5 mm 304 acceptable). Deburr.
- Tolerances: laser ±0.2 mm, bends ±0.5 mm.

Fitted by the customer — do not insert

- M3 rivnuts (Ø5 holes) in the body lips; solid rivets Ø3 (Ø3.2 holes) in the front corner tabs.
- M4 rivnuts (Ø6 grid) in the base plates as needed; M4 standoffs, rubber feet, panels, fans.

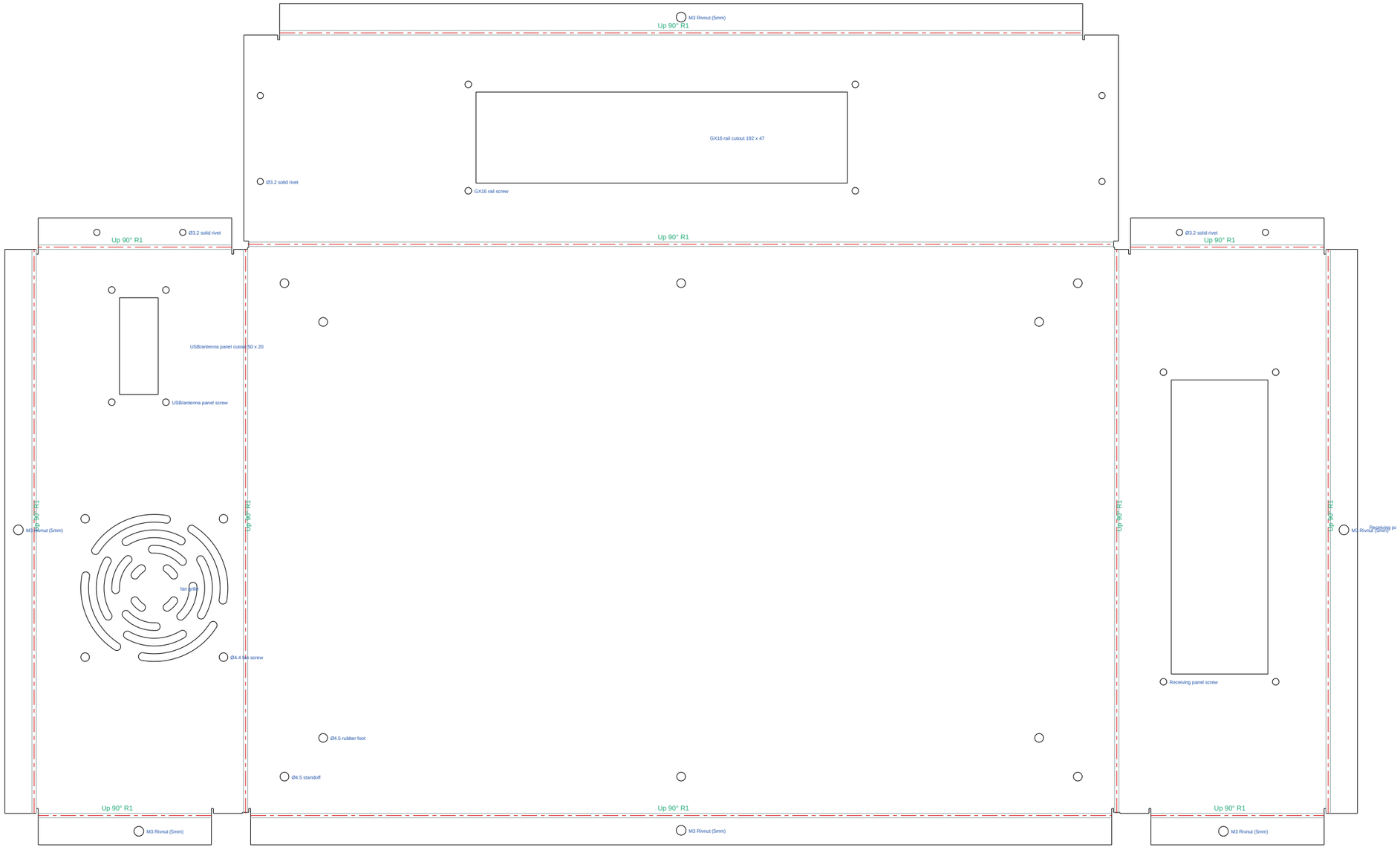
Checks: 642 design checks passed before these files were written — holes clear of bends (≥ 5 mm), webs ≥ 1.5 mm, folded parts free of clashes, folded flanges at least 2 mm apart wherever they are not meant to touch, whole assembly (fans, panels, rivnuts, base plate, standoffs) clear by 2 mm, lid clear of the body, every lid hole over its rivnut, base-plate standoffs over the floor's, panel cutouts and screw patterns matched to the printed panels. The files were then read back and re-folded from their own bend lines:

- refold body width: 452.00 (design 452.00)
- refold left wall height: 111.00 (design 111.00)
- refold right wall height: 111.00 (design 111.00)
- refold body depth (base): 297.00 (design 297.00)
- refold front wall height: 111.00 (design 111.00)
- refold lid width: 456.00 (design 456.00)
- refold lid depth: 300.00 (design 300.00)
- refold body width: 452.00 (design 452.00)
- refold left wall height: 171.00 (design 171.00)
- refold right wall height: 171.00 (design 171.00)

- refold body depth (base): 297.00 (design 297.00)
- refold front wall height: 171.00 (design 171.00)
- refold lid width: 456.00 (design 456.00)
- refold lid depth: 300.00 (design 300.00)

1 · CONTROL BOX · BODY

`control_body.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 699.0 × 434.8 mm · folded 452 W × 297 D × 111 H



CONTROL BOX - BODY

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

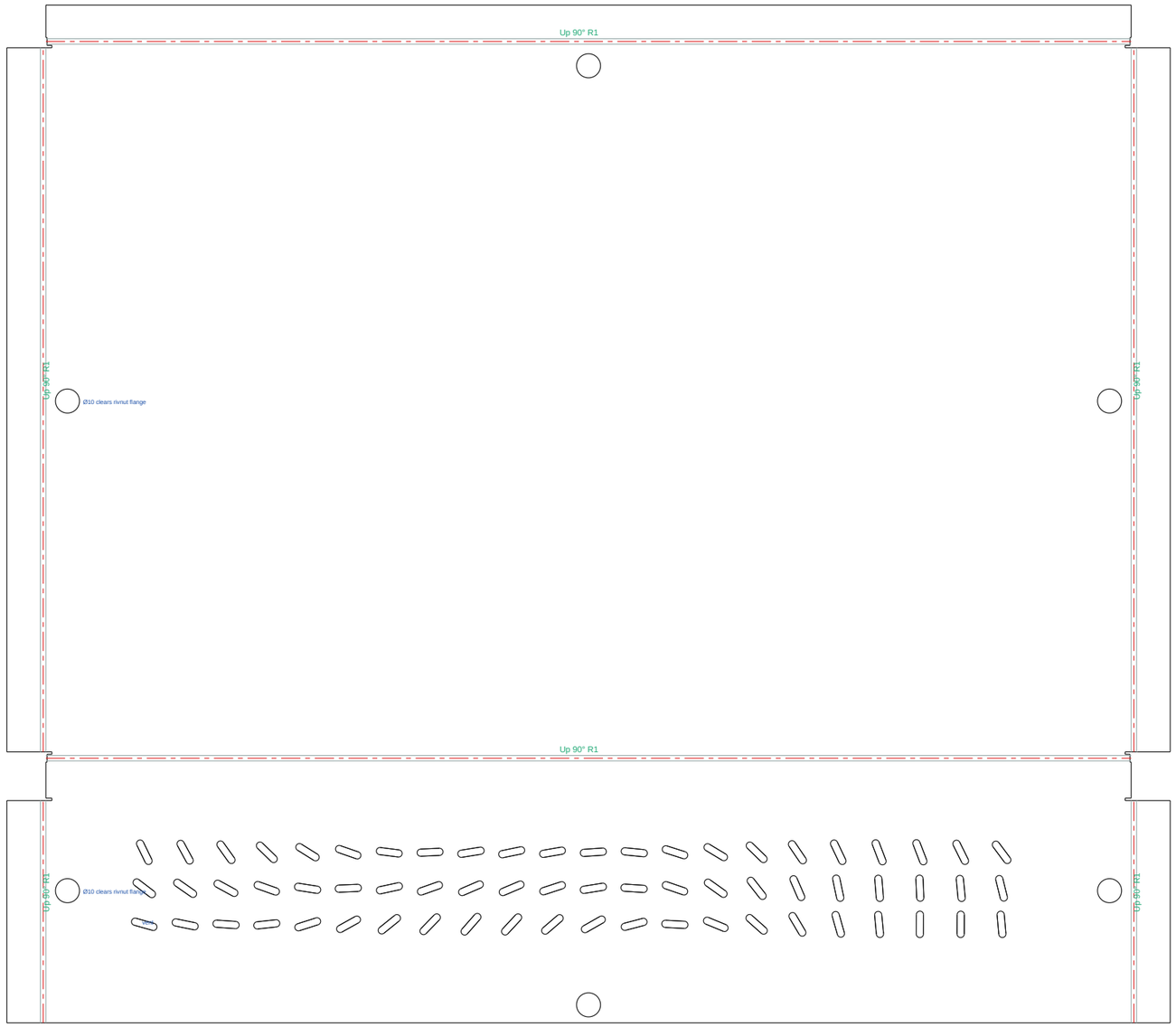
Folded outside: 452 W x 297 D x 111 H (body, lips in)

Inside: 450 x 295 x 110

Flat size: 699.0 x 434.8

2 · CONTROL BOX · LID

`control_lid.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 484.5 × 423.5 mm · folded 456 W × 300.0 D × 111 H (back)



CONTROL BOX - LID

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

Folded outside: 456 W x 300.0 D x 111 H at the back

Fits over the body with 1.0 mm per side inside the return lips

Flat size: 484.5 x 423.5

3 · CONTROL BOX · BASE PLATE

`control_base_plate.dxf` · 2.0 mm aluminium 5052 (or 1.5 mm 304) · qty 1 · flat 440.0 × 285.0 mm · folded flat, 440 × 285, notched



CONTROL BOX - BASE PLATE

Material: 2.0 mm aluminium 5052 (or 1.5 mm 304) Qty: 1 Units: mm

Flat part, no bends.

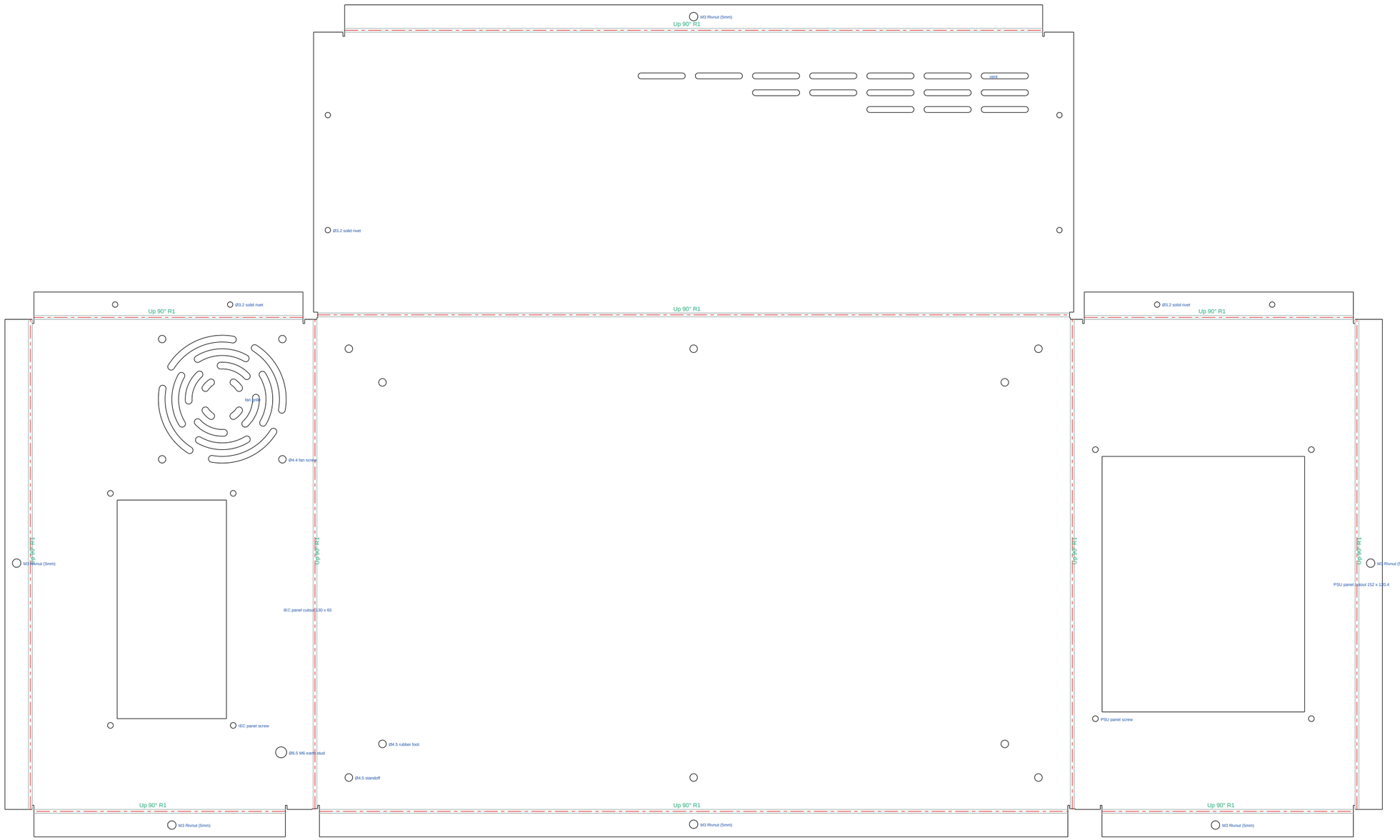
Flat, no bends. Sits on 6 x M4 standoffs, 5 mm.

Grid: 40 mm rows, every other row shifted 20 mm

Flat size: 440.0 x 285.0

4 · POWER BOX · BODY

`power_body.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 819.0 × 494.8 mm · folded 452 W × 297 D × 171 H



POWER BOX - BODY

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

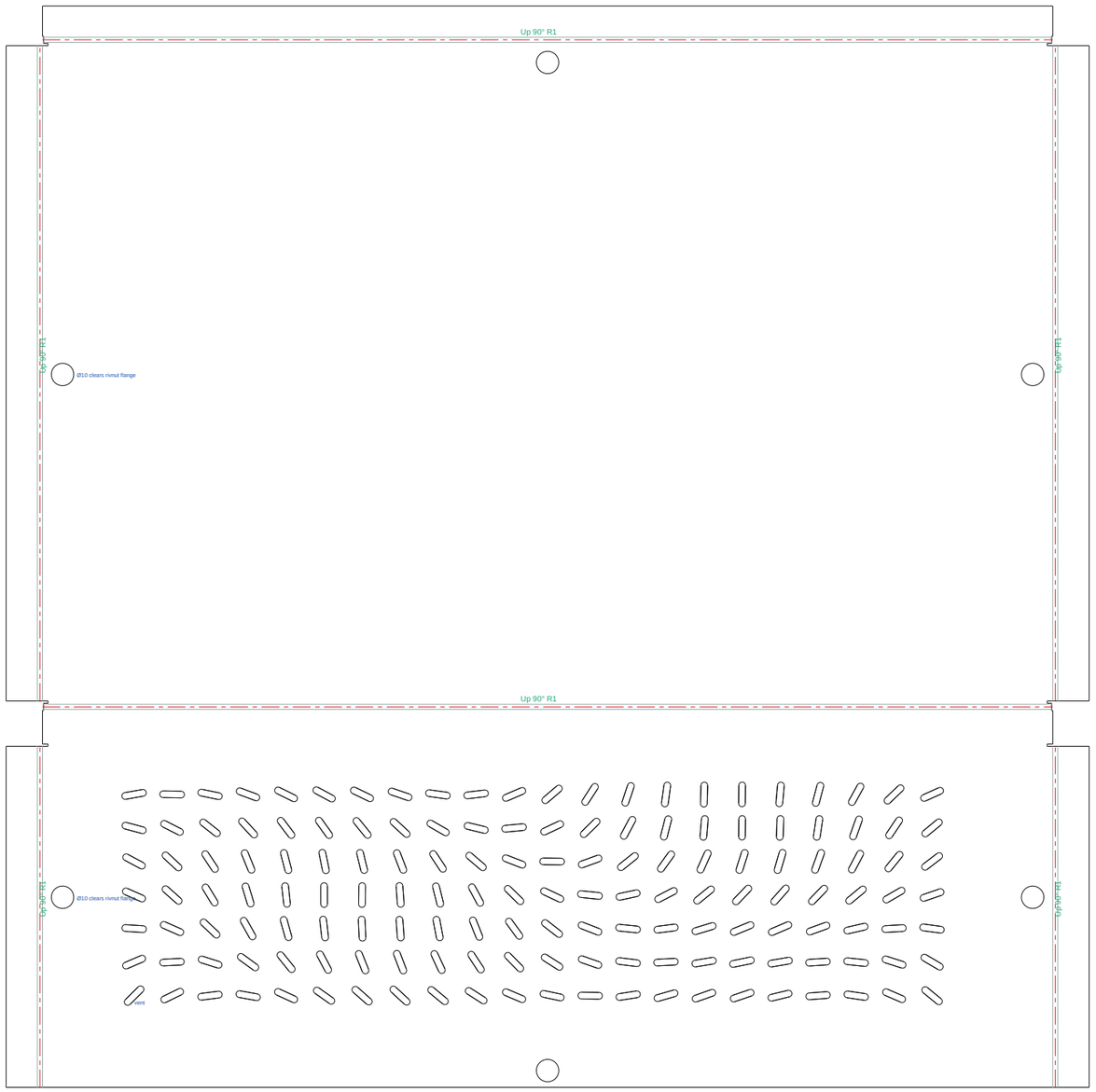
Folded outside: 452 W x 297 D x 171 H (body, lips in)

Inside: 450 x 295 x 170

Flat size: 819.0 x 494.8

5 · POWER BOX · LID

`power_lid.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 484.5 × 483.5 mm · folded 456 W × 300.0 D × 171 H (back)



POWER BOX - LID

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

Folded outside: 456 W x 300.0 D x 171 H at the back

Fits over the body with 1.0 mm per side inside the return lips

Flat size: 484.5 x 483.5

6 · POWER BOX · BASE PLATE

`power_base_plate.dxf` · 2.0 mm aluminium 5052 (or 1.5 mm 304) · qty 1 · flat 440.0 × 285.0 mm · folded flat, 440 × 285, notched



POWER BOX - BASE PLATE

Material: 2.0 mm aluminium 5052 (or 1.5 mm 304) Qty: 1 Units: mm

Flat part, no bends.

Flat, no bends. Sits on 6 x M4 standoffs, 5 mm.

Grid: 40 mm rows, every other row shifted 20 mm

Flat size: 440.0 x 285.0